

SAVEETHA SCHOOL OF ENGINEERING
CSA 15- CLOUD COMPUTING AND BIG DATA ANALYTICS
LIST OF EXPERIMENT 14

14. Demonstrate Infrastructure as a Service (IaaS) by creating a Virtual Machine using a Public Cloud Service Provider (Azure), configure with required memory and CPU

Aim

To demonstrate Infrastructure as a Service (IaaS) by creating and configuring a Virtual Machine (VM) using Microsoft Azure with the required CPU, memory, storage, and networking resources.

Procedure

1. Sign in to the Microsoft Azure Portal.
2. Select Virtual Machines from the Azure services.
3. Click Create → Azure Virtual Machine.
4. Select the Subscription and create/select a Resource Group.
5. Enter a suitable Virtual Machine Name and select the required Region.
6. Choose the required Operating System Image, such as Windows Server or Ubuntu Linux.
7. Under Size, select a VM configuration with the required CPU (vCPUs) and Memory (RAM).
8. Configure the Administrator Account with a username and password or SSH key.
9. Configure the required Disk/Storage settings.
10. Configure Networking by selecting the Virtual Network, Subnet, Public IP, and Network Security Group.
11. Allow the required inbound port, such as RDP (3389) for Windows or SSH (22) for Linux.
12. Click Review + Create and wait for Azure validation.
13. Click Create to deploy the Virtual Machine.
14. After deployment, open the VM and verify that its status is Running.
15. Connect to the VM using Remote Desktop for Windows or SSH for Linux.
16. Verify that the configured CPU, memory, storage, and operating system are available.

Result

A Virtual Machine was successfully created and configured on Microsoft Azure with the required CPU, memory, storage, and networking resources. Thus, the Infrastructure as a Service (IaaS) model was successfully demonstrated using Azure.

Screenshot:

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exp13 Virtual machine

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Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Monitor
- Resource visualizer
- Connect
- Networking
- Settings
- Availability + scale
- Security
- Backup + disaster recovery

Virtual machine

Computer name	exp13
Operating system	Windows (Windows Server 2025 Datacenter Azure Edition)
VM generation	V2
VM architecture	x64
Agent status	Ready
Agent version	2.7.41491.1216
Hibernation	Disabled
Host group	-
Host	-
Proximity placement group	-
Colocation status	N/A
Capacity reservation group	-

Networking

Public IP address	20.40.49.163 (Network interface exp13367)
	1 associated public IPs
Public IP address (IPv6)	-
Private IP address	172.16.0.4
Private IP address (IPv6)	-
Virtual network/subnet	vnet-centralindia-1/snet-centralindia-1
DNS name	Configure

Size

Size	Standard B2as v2
vCPUs	2
RAM	8 GiB
Threads per core	2

Source image details

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Virtual machine

Disk controller type SCSI

Azure Spot

Azure Spot	-
Azure Spot eviction policy	-

Availability + scaling

Availability zone (edit)	1
Extended zone	-
Availability set	-
Scale Set (attach)	-

Security

Security type	Trusted launch
Enable secure boot	Enabled
Enable vTPM	Enabled

Source image details

Source image publisher	MicrosoftWindowsServer
Source image offer	WindowsServer
Source image plan	2025-datacenter-azure-edition

Disk

OS disk	exp13_OsDisk_1_604b92f0ec934cb798c66af3115321de
Encryption at host	Disabled
Azure disk encryption	Not enabled
Ephemeral OS disk	N/A
Data disks	0

Auto-shutdown

Auto-shutdown	Not enabled
Scheduled shutdown	-

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